

5G Mobile Communications Systems (5GMCS)

Development of a use case of the 5G
technology

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Mobile Communications Research Group (GRCM)

Course 2026/27
Fall Term

Objectives

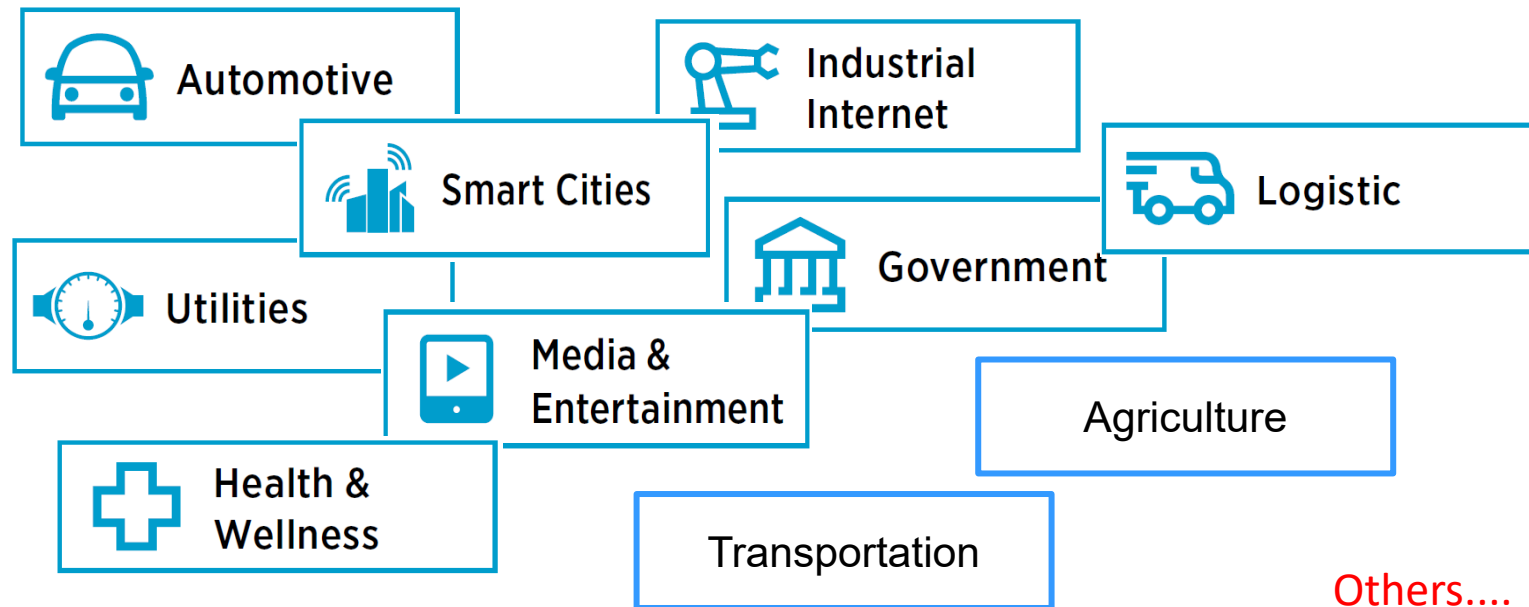
- Work on the elaboration of a use case of the 5G technology
- To be done in teams of 4 students
- Evaluation (30% of total course mark) will be based on:
 - 1) Oral presentations**
 - 2) Written report (Template available in Atenea)**

Expected contents

1) Description of the needs and innovation opportunity of applying 5G in a given sector

- Each team needs to select one use case in a given sector:

– Some examples of sectors:

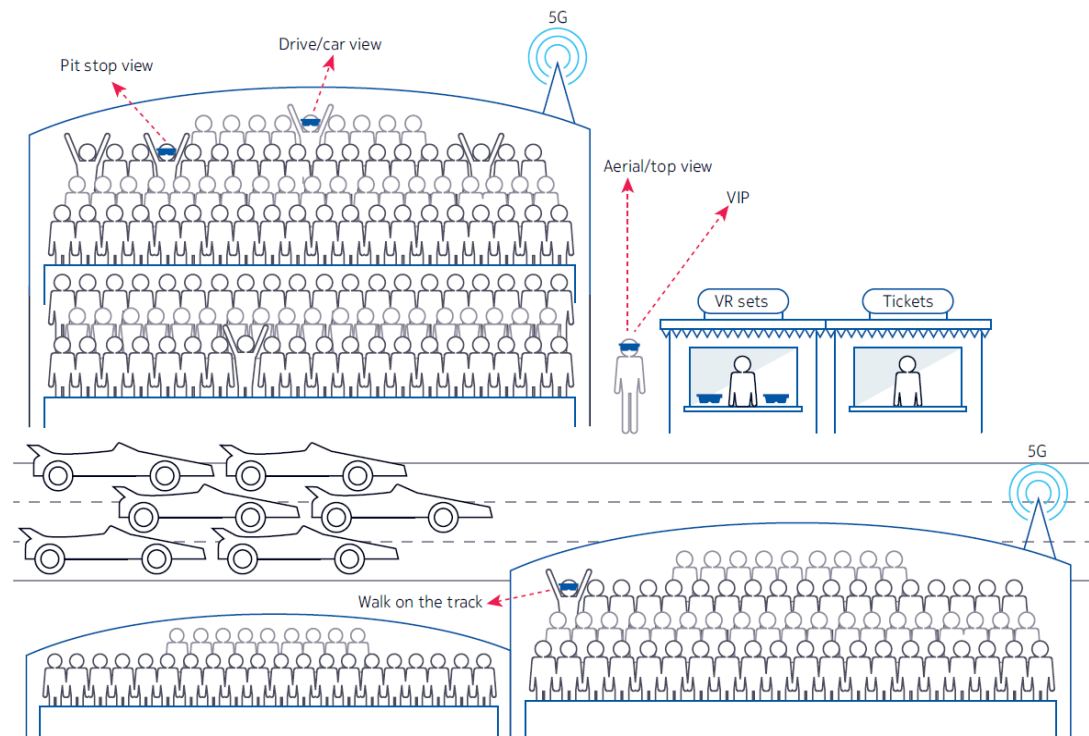


Expected contents

1) Description of the needs and innovation opportunity of applying 5G in a given sector

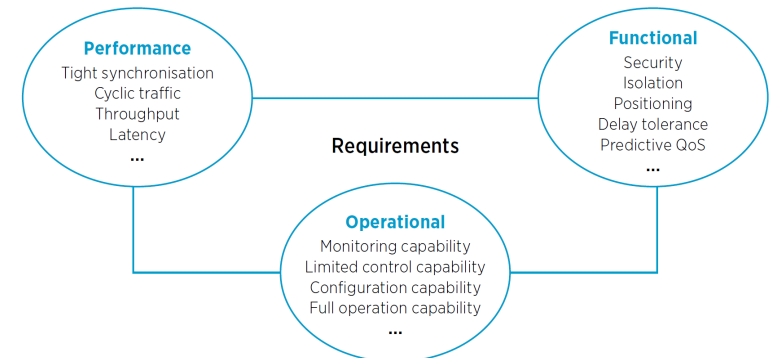
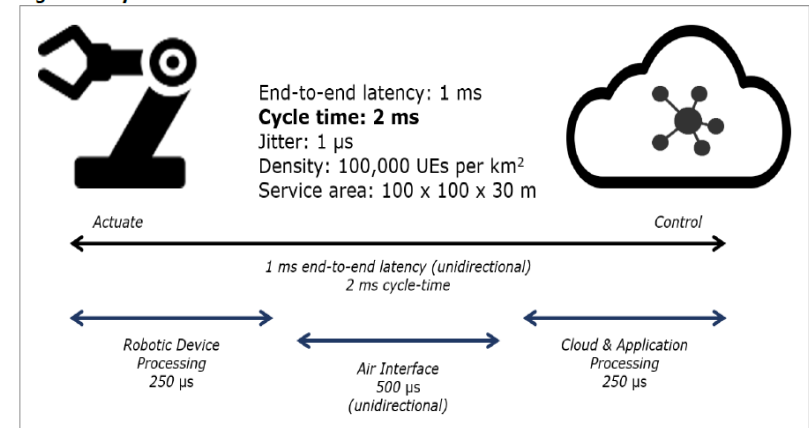
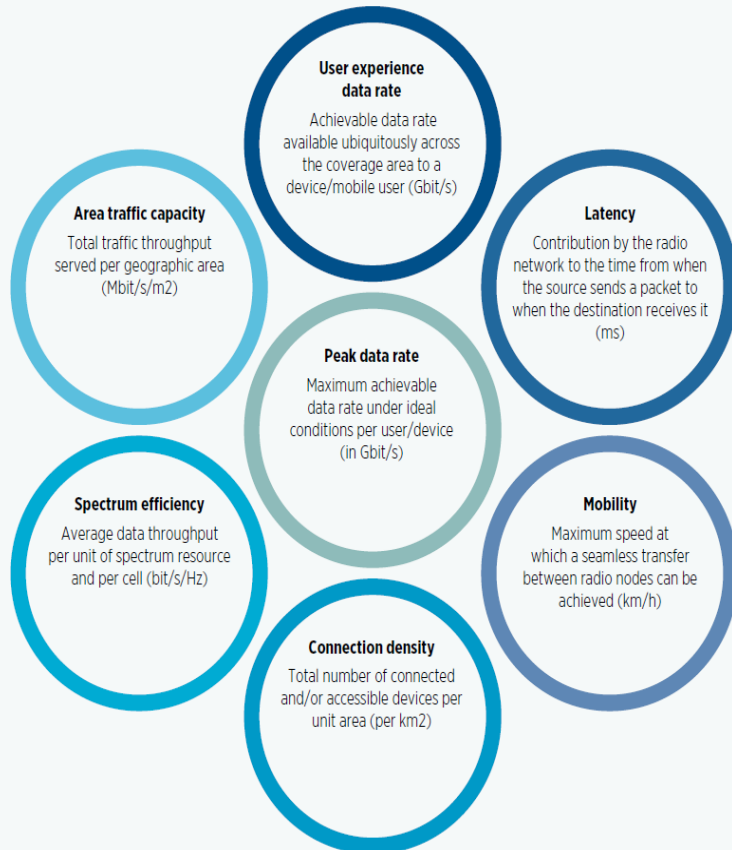
- Describe the use case in the selected sector:
 - Actors involved in the use case (e.g. operators, end users, venue owner, etc.)
 - Role played by each one
 - Expected improvements/benefits

Example: “Virtual Reality / Augmented Reality in sport events”



Expected contents

2) Description of technical requirements and required capabilities



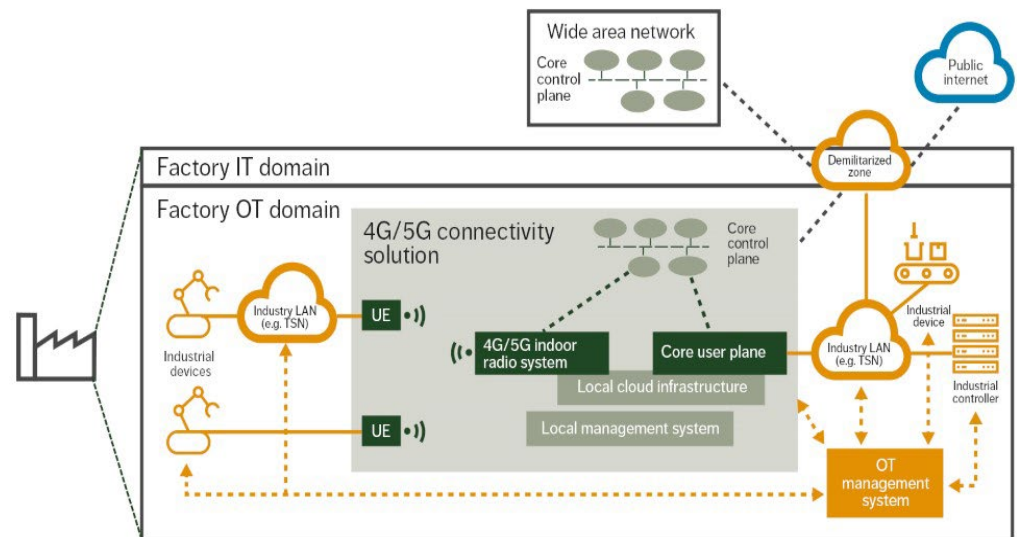
Expected contents

3) Design aspects

- Technical considerations for realizing the use case using 5G technologies. Examples:
 - Involved user equipment (smartphones, IoT devices, accesories, drones, sensors, etc.)
 - Network equipment (antennas, servers, edge computing, etc.)
 - Network architecture (diagram)
 - Spectrum, operation frequencies
 - Integration with existing systems
 - Operation and management of the system
 - Phases of deployment and implementation
 - ...

Example: manufacturing solution architecture

5G connectivity solution for the factory shop floor consisting of a local, on-premises 4G/5G connectivity solution that uses licensed spectrum. Integrated with mobile operator-provided wide area network.



Expected contents

4) Techno-economic assessment and viability

- Techno-economic aspects:
 - Deployment and operation costs
 - Business model
 - New revenues
 - Cost reductions
 - Investment return
- Regulatory aspects
- Critical assessment:
 - Strengths, Weaknesses, Opportunities, Threats (SWOT) analysis

Expected contents

5) References

- Sources of information that have been used to sustain the analysis:
 - Technical specifications
 - Market prospect studies
 - Documents from industry associations
 - Technical papers
 - ...

Formatting guidelines:

- Format of references must be complete:
 - [1] Authors, "Title", publication (journal, conference, etc.), year, link.
 - [2] Authors, "Title", publication (journal, conference, etc.), year, link.
 - ...
- References must be cited in the text, e.g. "...this value is taken from [1]", "Figure X, based on [2] shows the architecture of"
- Sources of figures and text extracted from other documents must be explicitly referenced.
- Sections/sub-sections must be numbered.

Initial materials

- Documents available in Atenea:
 - Template of the report
 - Some reference documents with examples of descriptions and analysis of sectors of application of 5G and use cases

Deadlines

- **September 22nd:** Presentation of the activity and expectations
- **September 27th:** Send e-mail to jordi.perez-romero@upc.edu indicating the **team composition**, the **selected sector** and, if already identified, the **selected use case**.
- **October 27th:** First oral presentation (max. 15 min)
 - Description of the use case, technical requirements and required capabilities
- **December 22nd:** Final oral presentation (max. 20 min)
 - Main focus on the design and techno-economic analysis. Only a brief introduction of use cases and requirements at the beginning.
- **December 23rd:** Delivery of the written report by e-mail to jordi.perez-romero@upc.edu

Follow-up meetings with the professor can be held during all the process. It is recommended that each group makes at least one meeting.